

Variant representation in RDF

BioHackathon series:

SWAT4HCLS BioHackathon
2025

Barcelona, Spain, 2025
[variant-rdf](#)

Submitted: 08 Aug 2026

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Published by [BioHackrXiv.org](#)

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Abstract

During the International SWAT4HCLS conference held on 24-27th February 2025 in Barcelona (Spain), we detected an emerging number of novel RDF models to represent variant information in genomic datasets potentially hindering data reuse. We tackled the question how semantic representations can enhance the interoperability of variant data for clinical applications. Here we report our initial results on genomic variant schema alignment.

Introduction

While variant representation is a cornerstone for understanding the molecular mechanisms underlying disease biology, there is no standard to represent this variation in RDF. During the International SWAT4HCLS conference held on 24-27th February 2025 in Barcelona (Spain), we detected an emerging number of novel RDF models to represent variant information in datasets. This challenges interoperability with existing Semantic Web based research infrastructures and hinders data reuse for example for variant interpretation. As part of the SWAT4HCLS BioHackathon 2025, we here report that we worked on the mapping and merging of different ontology models about genetic variants identified during the conference.

Results

Identifying variant RDF models

From talks and posters presented during the conference, we identified several projects devoted to representing genomic variants (see the list in the [‘Summary’ sheet](#)). According to diverse driving use cases, from human health to plant sciences, each project presented a different case-tailored and ontology-based RDF model.

Aligning existing RDF models

Our approach was to use the GA4GH VRS RDF schema (Queralt-Rosinach et al., 2026) as the standard. We leveraged the updated GA4GH VRS RDF Schema to align the different models and prototype a pipeline for annotating variant models mappings to VRS in RDF. See our mappings in TSV format in a dedicated [‘RDF model’ sheet](#)). Here, we depict an example of our mapping work for the SemBeacon RDF model:



Figure 1: Illustration of the SemBeacon RDF model mapping to GA4GH VRS RDF model.

Converting alignments into SSSOM standard

The final step in our alignment work was to prototype a pipeline to structure our semantic mappings using the SSSOM standard (Matentzoglou et al., 2022) in order to be reusable by the Semantic Web community. SSSOM provides a standard way to describe a lot of rich information about mappings and comes with a [Python toolkit](#) to translate a SSSOM compliant TSV file into OWL reified axioms (see the SSSOM annotation pipeline on GitHub).

Discussion

At the SWAT4HCLS conference, we detected the proliferation of different RDF models to represent variant information, identified their commonalities and disjointnesses, and defined a strategy to enable their interoperability. Our agreed strategy was to align all models to a common core model that acts as a reference and collects all shared features. As a best practice in the field, we utilised existing standards to map and share our manually derived mappings. Even though every model is driven by a particular use case that confers unique terms and features, we observed that all models have overlap with the GA4GH VRS RDF model. Therefore, we selected this model as our core model since GA4GH VRS (Wagner et al., 2021) is a global standard and, moreover, its design is modular and is based on object-oriented engineering to enable granularity and flexibility in representing molecular variation from basic to specific concepts. Another outcome was that we created a Python tool publicly available on Github [link] to describe and share with the community our curated mappings with the SSSOM standard. Finally, we want to highlight that the BioHackathon along the conference set an ideal opportunity for multiple groups to discuss how they had represented variation data in RDF and to define together a solution to enable interoperability.

Impact

SWAT4HCLS community: Strengthened the SWAT4HCLS (Semantic Web Applications and Tools for Healthcare and Life Sciences) community by sharing our RDF-based solutions, encouraging adoption of semantic standards in genomics.

Acknowledgements

We thank the organizers of the DBCLS BioHackathon 2025 for providing the venue and support for this work. We also acknowledge the developers of GA4GH VRS for providing the foundational model that made this work possible.

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Author contributions

